1. Consider the following context-free grammar (CFG) G:

$$
\begin{aligned}
E &\rightarrow E + T \mid T \\
T &\rightarrow T \times F \mid F \\
F &\rightarrow (E) \mid \mathtt{c}
\end{aligned}
$$

Give parse trees and derivations for each of the following strings. (10 points)

(a) c
(b) c+c
(c) c+c×c
(d) (c+c)×c

Matt Kruger HW 4

$E \rightarrow E + T \mid T$

$T \rightarrow T \times F \mid F$

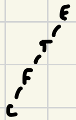
$F \rightarrow (E) \mid c$

a) $E \overset{*}{\Rightarrow} c$

derivation:

$$E \Rightarrow T \Rightarrow F \Rightarrow c$$

Tree:

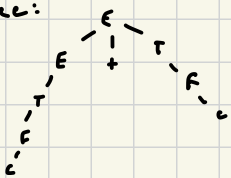


b) $E \overset{*}{\Rightarrow} c + c$

derivation:

$$E \Rightarrow E + T \Rightarrow T + T \Rightarrow F + T \Rightarrow c + T \Rightarrow c + F \Rightarrow c + c$$

Tree:



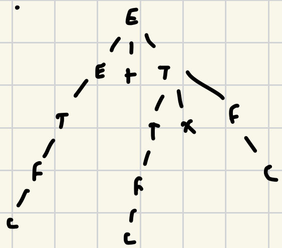
c) $E \overset{*}{\Rightarrow} c + c \times c$

derivation:

$$E \Rightarrow E + T \Rightarrow T + T \Rightarrow F + T \Rightarrow c + T \Rightarrow c + T \times F \Rightarrow c + F \times F \ \ldots$$

$$\ldots \Rightarrow c + c + F \Rightarrow c + c \times c$$

Tree:



1.d is on next page →

d) $E \overset{*}{\Longrightarrow} (c+c) \times c$

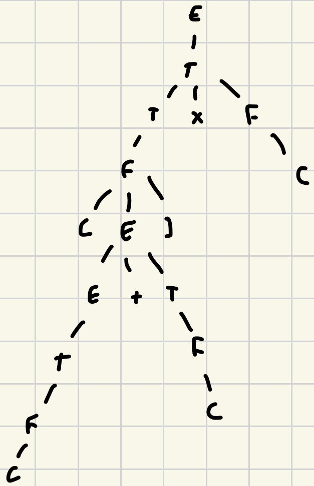
derivation:

$E \Longrightarrow T \Longrightarrow T \times F \Longrightarrow F \times F \Longrightarrow (E) \times F \Longrightarrow (E+T) \times F \Longrightarrow (T+T) \times F$ ...

... $\Longrightarrow (F+T) \times F \Longrightarrow (c+T) \times F \Longrightarrow (c+F) \times F \Longrightarrow (c+c) \times F$ ...

... $\Longrightarrow (c+c) \times c$

Tree:



2. For each of the following languages, give a context-free grammar that generates it. In all parts, the alphabet is $\Sigma = \{0, 1\}$. (12 points)

(a) $\{w \in \Sigma^* \mid w \text{ starts and ends with the same symbol}\}$
(b) $\{w \in \Sigma^* \mid \text{ the length of } w \text{ is odd}\}$
(c) $\{w \in \Sigma^* \mid w \text{ is a palindrome}\}$
(d) The empty set

A palindrome is a string that is the same as its reverse. For example, 10001 and 0110 are palindromes, but 100 and 1100 are not.

a) w starts and ends with the same symbol

$S \rightarrow 0A0 \mid 1A1 \mid 0 \mid 1$
$A \rightarrow 0A \mid 1A \mid \varepsilon$

b) $|w|$ is odd

$S \rightarrow 0S0 \mid 0S1 \mid 1S0 \mid 1S1 \mid 0 \mid 1$

c) palindrome

$S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1$

d) $\emptyset$

unsure about this one...

$S \rightarrow A$
$A \rightarrow S$

3. Consider the language

$$A = \{a^i b^j c^k \mid i = j \text{ or } j = k, \text{ where } i, j, k \geq 0\}$$

(a) Give a context-free grammar that generates A. (6 points)

(b) Is your grammar ambiguous? Why or why not? (4 points)

a) CFG:

$S \to S_1 \mid S_2$
$S_1 \to DC$
$D \to aDb \mid \varepsilon$
$C \to cC \mid \varepsilon$
$S_2 \to AB$
$A \to aA \mid \varepsilon$
$B \to bBc \mid \varepsilon$

b) yes, this language is ambiguous as there exists accepted strings with multiple left-most derivation paths

example: generating string abc

$S \Rightarrow S_1 \Rightarrow DC \Rightarrow aDbC \Rightarrow abC \Rightarrow abc$

$S \Rightarrow S_2 \Rightarrow AB \Rightarrow aB \Rightarrow abBc \Rightarrow abc$

4. For each of the following languages, give an informal description and a state diagram of a pushdown automaton (PDA) recognizing that language. Design the PDA directly, and not by converting a CFG into a PDA. In all parts, the alphabet is $\Sigma = \{0, 1\}$. (12 points)

(a) $\{w \in \Sigma^* \mid w \text{ starts and ends with the same symbol}\}$

$\boxed{4a}$

(b) $\{w \in \Sigma^* \mid \text{the length of } w \text{ is odd}\}$

(c) $\{w \in \Sigma^* \mid w \text{ is a palindrome}\}$

(d) The empty set

a) $\{ w \in \Sigma^* \mid w$ starts and ends with the same symbol $\}$

Q: $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}\}$

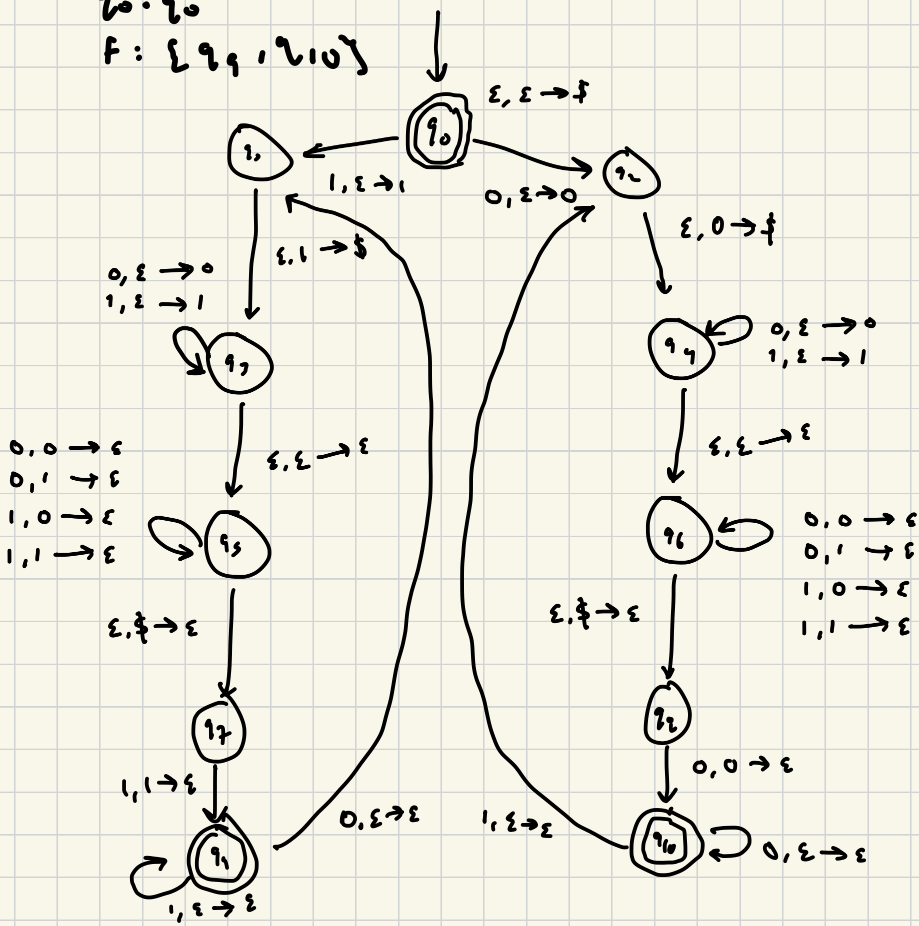
Σ: $\{0, 1\}$

Γ: $\{0, 1, \$\}$

δ: not showing; "informal description"

q_0: q_0

F: $\{q_9, q_{10}\}$



4 b)

b) $\{ w \in \Sigma^* \mid$ the length of w is odd $\}$

$Q: \{q_0, q_1, q_2, q_3\}$

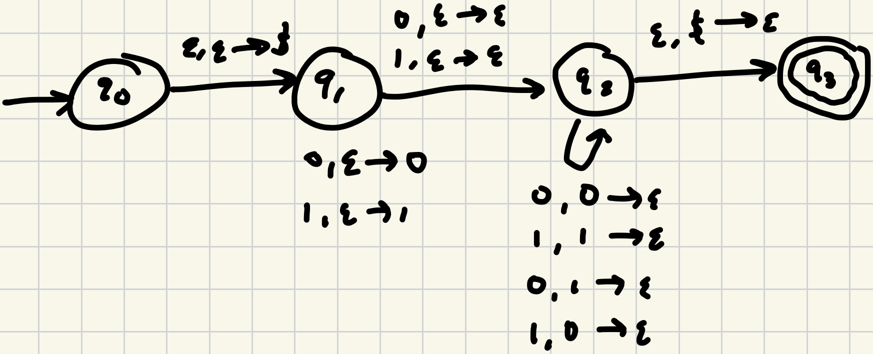
$\Sigma: \{0, 1\}$

$\Gamma: \{0, 1, \$\}$

δ: Not showing; "informal"

$q_0: q_0$

$F: \{q_3\}$



$\boxed{4c}$

c) $\{ w \in \Sigma^* \mid w \text{ is a palindrome} \}$

i.e. $w = w^R$

Q: $\{q_0, q_1, q_2, q_3\}$

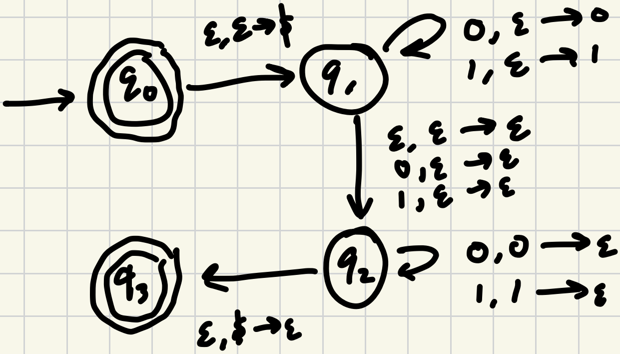
Σ: $\{0, 1\}$

Γ: $\{0, 1, \$\}$

δ: not showing; "informal"

q_0: q_0

F: $\{q_0, q_3\}$



$\boxed{4d}$

d) $\emptyset$

Q: $\{q_0\}$

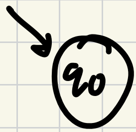
Σ: $\{0,1\}$

Γ: $\emptyset$

δ: $\emptyset$

q_0: q_0

F: $\emptyset$



5. Give an informal description and a state diagram of a pushdown automaton recognizing the language

$$A = \{\mathtt{a}^i\mathtt{b}^j\mathtt{c}^k \mid i = j \text{ or } j = k, \text{ where } i, j, k \geq 0\}$$

Once again, design the PDA directly. (6 points)

Q: $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7\}$

Σ: $\{a, b, c\}$

Γ: $\{a, b, \$\}$

δ: not showing; "informal"

q_0: q_0

F: $\{q_5, q_7\}$